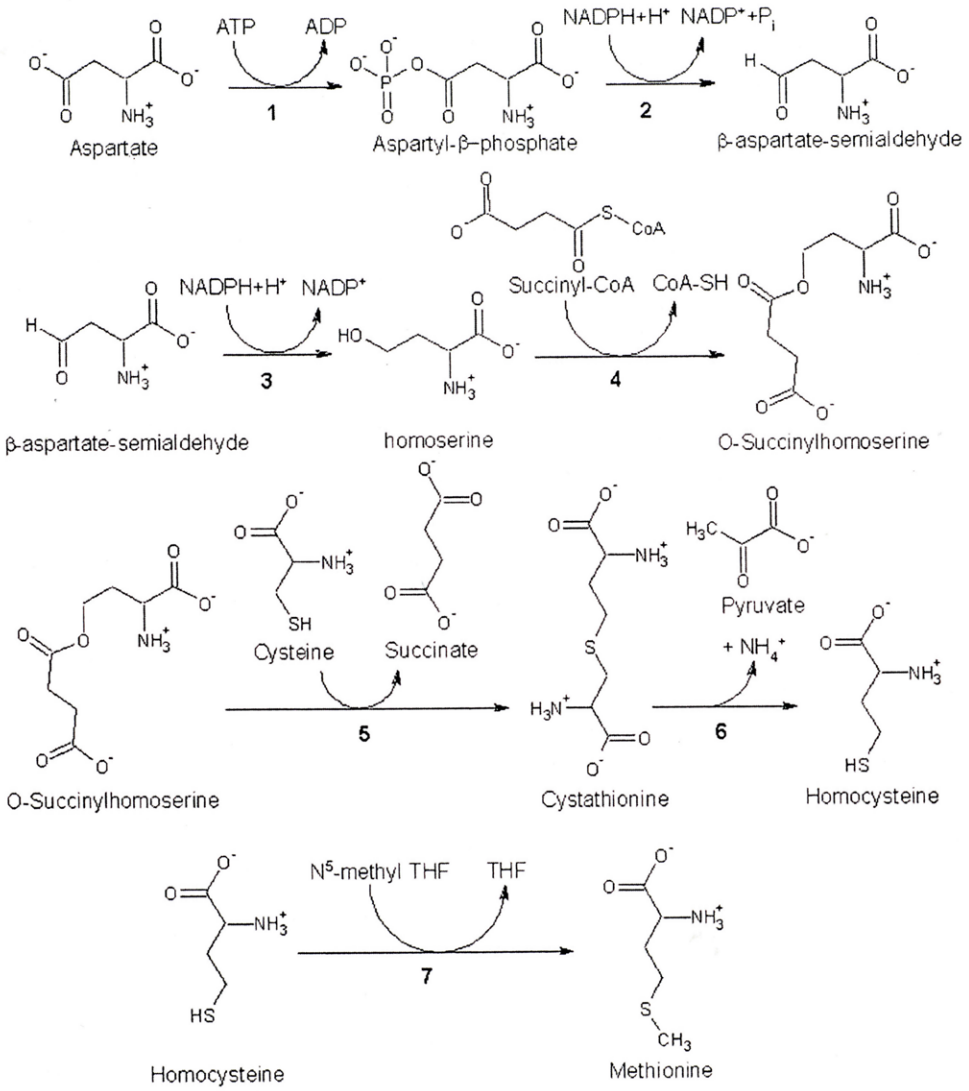
Biochemistry Cume

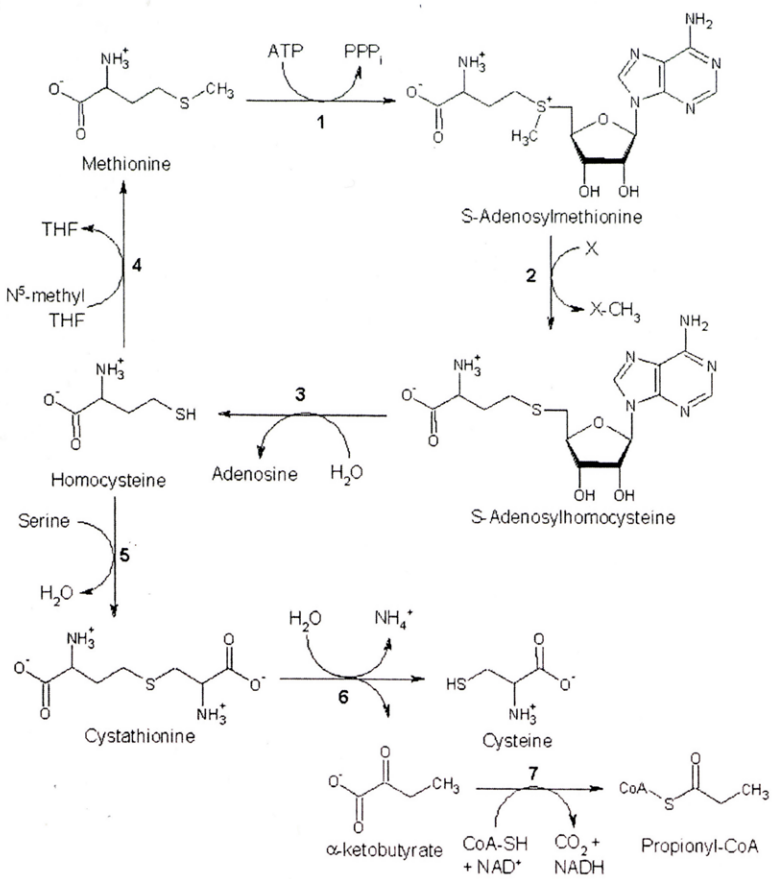
Total points: 100

1. Write the reaction for the transamination of an amino acid in terms of the Cleland notation **(20 points).**

2. Given below are the biosynthesis and degradation pathways of methionine.



Biosynthesis pathway of Methionine



Degradation Pathway of methionine

In the above given biosynthesis pathway, methionine is synthesized by the methylation of homocysteine in a reaction in which N5-methyl-THF is the methyl donor. Yet, in the breakdown of methionine, its demethylation occurs in three steps in which SAM (S-adenosyl methionine) is an intermediate. Explain why this reaction does not occur via a simpler one-step reversal of the methylation reaction. (**35 points**)

3. A student on a particular diet expends 10,000 kJ.day^{-1} while excreting 40 grams of urea. Assuming that protein is 16% nitrogen by weight and the metabolism yields 18 kJ.mol^{-1}, what percentage of the student's energy requirement is met by protein. (**30 points**).

4. Why are phenylketonurics warned against consuming products containing the artificial sweetner aspartame (**15 points**)